

ST THOMAS CYRIL KING, Virgin Islands, U.S.

WMO: 785430

Lat: 18.336N

Lon: 64.980W

Elev: 20

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 98-19

WBAN: 11640

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	70.5	71.7	57.5	70.5	76.2	60.2	78.0	76.5	22.8	78.9	21.2	79.0	6.7	80	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	89.8	78.2	88.9	78.1	88.2	78.0	80.9	85.9	80.3	85.7	79.9	85.5	14.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	151.6	83.7	78.7	149.1	83.6	78.1	146.0	83.6	44.5	86.5	43.9	85.7	43.3	85.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.8	17.6	DB	67.3	92.7	3.5	2.4	64.8	94.4	62.7	95.8	60.8	97.2	58.3	98.9	(4)
(5)			WB	58.4	81.6	5.6	1.2	54.3	82.4	51.0	83.1	47.8	83.8	43.7	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	78.0	77.9	78.2	79.7	81.5	83.3	83.8	83.8	83.7	82.7	80.9	79.3
	DBStd	2.84	1.60	1.44	1.96	1.71	1.79	1.61	1.65	1.74	1.68	1.82	1.69	1.69
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11344	869	782	874	890	976	999	1046	1049	1011	1013	926	909
	CDD65	5869	404	362	409	440	511	549	581	584	561	548	476	444
	CDH74	59383	2794	2396	2982	3925	5402	6485	6943	7083	6765	6233	4725	3650
	CDH80	17251	367	313	433	759	1453	2301	2593	2722	2556	2030	1099	625
Wind	WSAvg	10.1	10.3	10.7	10.3	10.3	10.4	11.0	11.4	10.6	8.8	8.4	9.0	10.5
Precipitation	PrecAvg	39.2	1.8	1.4	2.2	2.6	3.9	2.9	2.6	3.9	4.7	5.5	4.6	3.1
	PrecMax	60.2	3.7	2.7	5.7	16.4	11.3	8.1	5.6	8.5	12.8	16.9	11.2	7.5
	PrecMin	23.2	0.4	0.5	0.4	0.2	0.4	0.4	0.7	0.8	1.7	1.4	0.8	0.6
	PrecStd	10.8	0.9	0.6	1.5	3.3	2.9	2.2	1.4	2.1	2.3	4.1	3.3	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.2	85.2	85.9	86.7	88.4	91.3	91.6	91.7	91.4	89.3	87.6	86.3
		MCWB	72.6	72.8	74.0	75.5	77.5	77.8	77.4	78.3	78.8	77.9	77.1	75.3
	2%	DB	84.1	84.0	84.6	85.5	87.1	88.8	89.5	89.5	89.2	88.1	86.5	85.3
		MCWB	72.8	72.6	73.8	75.3	77.0	77.7	77.8	78.4	78.8	78.0	76.9	75.1
	5%	DB	83.3	83.2	83.7	84.7	86.2	87.8	88.4	88.5	88.2	87.3	85.8	84.5
		MCWB	72.8	72.6	73.5	75.1	76.7	77.7	77.6	78.3	78.7	77.9	76.8	74.8
	10%	DB	82.6	82.5	82.8	83.9	85.3	87.0	87.5	87.6	87.5	86.5	85.1	83.6
		MCWB	72.9	72.5	73.2	74.8	76.5	77.5	77.6	78.3	78.5	77.8	76.6	74.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	76.2	76.6	79.1	80.0	80.5	81.0	81.6	81.7	81.6	80.5	78.2
		MCDB	81.2	81.9	82.0	84.0	85.5	86.8	85.8	86.1	86.4	85.6	84.0	83.0
	2%	WB	75.4	74.8	75.8	77.9	79.0	79.7	80.1	80.7	80.8	80.6	79.2	77.0
		MCDB	80.6	80.8	81.4	83.0	84.8	86.3	85.8	86.3	86.0	84.8	83.4	82.3
	5%	WB	74.7	74.0	75.1	76.8	78.2	79.0	79.5	80.0	80.1	79.7	78.3	76.1
		MCDB	80.1	80.3	80.8	82.3	84.1	85.8	85.6	85.9	85.7	84.7	83.0	81.6
	10%	WB	74.0	73.4	74.4	75.8	77.4	78.3	78.9	79.5	79.6	78.9	77.6	75.4
		MCDB	79.7	79.9	80.3	81.7	83.4	85.1	85.3	85.6	85.5	84.7	82.6	81.1
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.1	8.7	8.5	7.9	8.2	8.9	8.7	8.4	8.3	8.6	8.7
		MCDBR	10.0	10.2	9.5	9.0	8.4	8.7	9.6	9.5	9.1	9.0	9.3	9.6
		MCWBR	4.3	4.3	4.0	4.0	3.6	3.6	3.8	4.0	3.9	3.9	4.2	4.3
	5% WB	MCDBR	8.8	9.1	8.2	7.6	7.5	8.1	8.7	8.7	8.3	7.7	8.5	8.4
MCWBR		4.4	4.4	4.1	4.0	3.7	3.8	4.0	4.1	4.2	4.2	4.4	4.2	
Clear-Sky Solar Irradiance	taub	0.371	0.382	0.383	0.407	0.447	0.515	0.522	0.484	0.440	0.393	0.384	0.376	
	taud	2.576	2.534	2.530	2.491	2.391	2.159	2.132	2.276	2.428	2.582	2.598	2.594	
	Ebn at Noon	290	292	294	285	271	251	249	261	273	285	284	285	
	Edh at Noon	30	33	34	36	39	48	50	43	37	31	29	29	
All-Sky Solar Radiation	RadAvg	1509	1762	1965	2076	2001	2014	2042	2008	1884	1652	1454	1400	
	RadStd	41	38	74	118	134	119	102	74	120	116	101	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+4.41	+0.58	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.41	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	+142	+144

Nomenclature: See separate page